

Why Michael Can't Spot The Attack: Evaluating User Perceptions and Awareness of Potential Public Wi-Fi Threats

Lauren Petrosian

lauren_petrosian@sfu.ca

Kathy Cho

kathy_cho@sfu.ca

Abstract— Public Wi-Fi is widely used in everyday life, but the attacks that target it such as Evil Twin, Packet Sniffing, MITM, SSL Stripping, and DNS Spoofing often go unnoticed by users. Security advice usually assumes that being aware of risks leads to safer behaviour, yet this assumption is rarely tested for specific attacks. We address this by conducting a two-part survey with 39 participants, separating how familiar they are with attack terminology from their ability to recognize and respond to real-world public Wi-Fi attack scenarios. Our study participants' responses are scored against an item-level rubric and where they are classified into four Wi-Fi threat awareness tiers named Unaware, Partially Aware, Mostly Aware, and Fully Aware. Our findings show that user awareness depends on the specific attack type and is not general. For example, 59% of participants were unaware of an SSL stripping attack, while most could identify DNS spoofing. Nearly half of the participants who considered themselves tech-savvy were only partially aware or less, and approximately one in four demonstrated optimism bias. The best predictor of awareness was whether participants had heard of the attack terms before, which suggests that attack-specific education is needed to help mitigate people's gaps in public Wi-Fi attack knowledge.

Keywords— *public Wi-Fi security, user awareness, scenario-based survey, threat perception, attack recognition*

I. INTRODUCTION

Public Wi-Fi is now a common aspect of everyday life, especially alongside the growth of using personal smart devices. Places such as Coffee shops, airports, libraries, hotels, universities and public transit systems routinely offer free wireless access, and users have grown accustomed to connecting to these networks for work, communication, entertainment and everyday tasks. The convenience of free unauthenticated Wi-Fi connection is very appealing, however this same convenience hides a range of security risks many users are unaware of [1].

Once connected to a public Wi-Fi network, a user is exposed to a variety of attacks where attackers can exploit the open and shared nature of these environments. While on the same network as a regular user, attackers can intercept unencrypted traffic, impersonate real Wi-Fi access points, downgrade secure connections to unprotected ones, or redirect users to malicious websites. These attacks in many cases are completely invisible to the victim. As such, victims of these attacks may browse the web, log into an account, or complete transactions with no idea that their data has been intercepted or manipulated [2].

These attacks are well documented within cybersecurity literature, and there are widely available tools to execute them. What is less understood is the user side of this situation. How much do public Wi-Fi users actually know about the specific threats they face, and does their knowledge align with the confidence they place in public networks? Security recommendations usually assume that user's awareness translates into safer behaviour, however this assumption has not been directly tested in the context of specific named attack types that users are likely to encounter [3].

This study investigates this gap by measuring public Wi-Fi threat awareness through a scenario-based survey. Instead of asking users whether they have heard of specific attacks, we describe realistic situations and assess if the participants can identify the threat and respond appropriately. By separating familiarity with terminology from practical recognition, our approach aims to capture the difference between what users know in theory and what they would actually do in practice. This study is guided by three research questions:

(RQ1): How aware are public Wi-Fi users of common security threats, and which specific attack types represent the greatest blind spots?

(RQ2): Does self-reported technical knowledge accurately predict actual threat awareness?

(RQ3): Do users exhibit optimism bias regarding public Wi-Fi security, where their concern does not match demonstrated awareness?

To answer these three questions, we conducted a two-part survey with 39 participants recruited from a range of professional and academic backgrounds. Part I consists of demographic and baseline behavioural data, which includes self-reported technical knowledge, Wi-Fi usage habits and security attitudes. Part II presented five scenario-based questions adapted from real-world public Wi-Fi attacks. The attacks include evil twin, passive packet sniffing, man-in-the-middle (MITM), SSL stripping and DNS spoofing. Participant responses were scored against a rubric designed to classify threat awareness into four tiers: Fully Aware, Mostly Aware, Partially Aware and Unaware. We used non-parametric statistical tests to compare the results across demographic segments, attack types and self-reported measures.

Our results show that public Wi-Fi threat awareness is uneven and attack-specific. The DNS spoofing scenario

received the highest scores among participants; however, 59% were classified as Unaware of SSL Stripping attacks, indicating a blind spot in awareness of protocol-layer attacks. Participants' self-reported technical knowledge is only a moderate predictor of actual threat awareness, where 47% of self-described tech-savvy participants scored at or below the Partially Aware tier. Almost one quarter (26%) of participants exhibited optimism bias, which is our measure that pairs low expressed security concern with a demonstrated inability to recognize common threats. The strongest predictor of awareness was not self-rated technical skill but earlier familiarity with attack terminology, suggesting that even minimal exposure to attack names and mechanisms can produce measurable gains in Wi-Fi threat awareness. This result helps us recommend targeted educational methods to address people's lack of Wi-Fi threat awareness.

This paper makes the following contributions:

1. We design a scenario-based survey instrument that measures practical public Wi-Fi threat awareness by separating familiarity with attack terminology from the ability to recognize and respond to attacks in context. This instrument covers five common attack types (Evil Twin, packet sniffing, MITM, SSL stripping, DNS spoofing) and classifies responses into four awareness tiers using an item-level rubric.
2. We conduct an empirical study with 39 participants and show that public Wi-Fi awareness is attack-specific rather than a general trait, with SSL Stripping as a near-universal blind spot and prior familiarity with attack terminology as the strongest predictor of overall awareness.
3. We identify and quantify optimism bias in public Wi-Fi use, finding that a meaningful minority of participants express low security concern despite being unable to recognize common threats.

The remainder of our paper is organized as follows. Section II discusses related literature to our study. Section III discusses background information on public Wi-Fi vulnerabilities and the specific attack types examined in this study. Section IV provides a proof-of-concept for the five attacks examined in this study to show that they are achievable by adversary with modest skills and available resources. Section V details our study's design and methodology. Section VI reports our study's results organized by research questions. Section VII discusses our results, limitations and directions for future work. Lastly, section VIII concludes this paper.

II. RELATED WORK

Public Wi-Fi security has been studied from three different research angles that are relevant to our work: attack feasibility, user behavior on public networks, and the relationship between awareness and protective behaviour.

Attack feasibility. There is a substantial amount of literature establishing that public Wi-Fi attacks are practical and low-cost to execute. Through Evil Twin attacks, rogue access points are easy to set up and hard for users to distinguish from legitimate ones [6, 17], and packet sniffing continues to be effective

against unencrypted network traffic even as HTTPS adoption has become more popular, specifically against DNS queries that expose browsing behaviour [7, 18]. MITM attacks on public networks can be done by ARP spoofing and using modern TLS interception tools [9, 19, 20], while protocol-layer threats such as SSL stripping and DNS cache poisoning attacks continue to be widespread as well [10, 21]. Vulnerabilities found at the WPA2 and 802.11 layers demonstrate that even networks that are compliant to standard are not inherently safe [16, 17]. These works show that the threat landscape we study is reproducible and attainable by adversaries.

User behaviour on public Wi-Fi. We researched how users interact with public networks, users continue to rely on unsecured Wi-Fi even when they report awareness of its risks [8], self-protective behaviours observed are inconsistent with user's expressed concern [29], and perceptions of "free" city-wide Wi-Fi diverge from its privacy implications [2, 5]. The previous studies assess awareness as a single construct, however, our contribution is to assess awareness by attacks type and show that it varies significantly.

Awareness and protective behaviour. Other literature around usable-security shows that targeted specific interventions outperform general advice [23, 26], where tailored education creates greater measurable gains for non-technical audiences [22, 27]. Additionally, prior frameworks to evaluate behaviours separate recognition, understanding, and behavioural responses, which our four-tier rubric mirrors. Our study extends from these works by comparing awareness versus behaviour specifically in the context of public Wi-Fi attacks.

III. BACKGROUND

A. Public Wi-Fi Security Overview

Public Wi-Fi networks operate fundamentally differently from private networks in ways that expose users to risk. Most public hotspots are open or use a shared password, meaning all users share the same pre-shared key (PSK), from which per-session encryption keys are derived through a 4-way handshake [4]. An attacker who knows the PSK and captures another user's handshake by passively listening or by forcing a de-authentication and re-association, can derive that user's session keys and decrypt their traffic [4]. Even networks that follow the IEEE 802.11 standard that supports security protocols including WPA and WPA2 leave users vulnerable in this shared-medium model [5].

The open and unverified nature of public Wi-Fi introduces users to additional risks beyond protocol-level weaknesses. Incoming users have no way of verifying whether the network is legitimate, whether other users on the same network are benign, or whether the infrastructure itself has been compromised. These weaknesses and uncertainties create an attractive environment where adversaries can carry out their attacks with minimal effort and limited technical resources.

B. Common public Wi-Fi Attacks

In this subsection, we describe the five attack types tested in our study. Each attack type is well documented in cybersecurity literature and can be carried out with widely available tools.

An Evil Twin (ET) attack involves an adversary setting up a rogue wireless access point that impersonates a legitimate network's name and appearance. For example, a coffee shop offers a free Wi-Fi network named "Cafe_Guest", an attacker can create an identically named network that has a stronger signal nearby to tempt a victim into connecting. After the victim connects to the illegitimate network, all the victim's traffic is then routed through the attacker's device, exposing the user's credentials and other sensitive data [6]. The reason why this attack is particularly effective is that most consumer devices are configured to automatically associate with previously used SSIDs and to prefer connecting to stronger signals. This means that a user can fall victim to an ET attack without any explicit action on their part.

Packet sniffing is the use of software tools to passively capture and analyze data packets transmitted over a network. When on an unencrypted public network, an attacker can use tools to capture traffic, including login credentials, emails, and other sensitive data passing through the network without actively interfering with connections [7]. Packet sniffing is also used for actual network diagnostics, however, it can be easily weaponized in public Wi-Fi environments to violate a user's privacy and security.

Man-in-the-Middle (MITM) attacks occur when an attacker secretly intercepts and potentially alters communication between two parties who believe they are communicating directly. While a user is on public Wi-Fi, an adversary on the same network can use tools to redirect traffic through their own device, allowing the adversary to eavesdrop on unencrypted data and capture sensitive information [8]. These attacks are dangerous because they are invisible to victims. A user may complete a transaction or log into a website without any indication that their data has been intercepted. Other attack techniques such as DNS spoofing and SSL stripping are types of MITM attacks, however, in this study we treat them separate scenarios since they test different attack recognition skills from the participants.

SSL stripping is a specialized MITM technique in which an adversary intercepts packets and downgrades a user's secure HTTPS session to an unprotected HTTP session without the user's knowledge [9]. With this attack, adversaries can exploit the fact that users typically type domains into the URL bar without the "https://" prefix, which causes the user's initial request to be sent over plaintext HTTP. And so, the adversary intercepts the server's HTTP-to-HTTPS redirect, maintains an HTTPS connection to the server on the user's behalf and serves HTTP to the user instead. Then from the user's perspective, the website appears to load properly, but the data exchanged during the session travels in plaintext to the attacker [9]. The adversary can now observe all browser-to-server traffic without breaking encryption, and can capture credentials and session data directly. HTTP Strict Transport Security (HSTS) was introduced to address this attack. HSTS instructs browsers to refuse plaintext connections to a domain once an HSTS policy has been received

[10]. However, SSL stripping remains as a feasible attack against sites not covered by HSTS.

The last attack covered in our study is DNS spoofing. This is an attack in which an attacker corrupts the process of domain name translation to IP addresses. Normally, when a user enters a website address into their browser, a DNS resolver looks up the corresponding IP address and directs the connection to the correct server. Under this attack, this process is manipulated and unknowingly directs the user to a malicious server controlled by the attacker [11]. This attack extends beyond credential theft, fraudulent destinations can distribute malware or serve as a gateway to further attacks. Because DNS was originally designed without built-in verification, it remains inherently vulnerable to manipulation and even though DNSSEC was developed to address this, DNS is still left vulnerable to attacks [12]. Specifically for public Wi-Fi threats, DNS-over-HTTPS (DoH) and DNS-over-TLS (DoT) encrypt queries between the user's device and resolver which prevents on-path attackers from tampering with responses [13, 14]. Both services are supported in modern browsers and operating systems, but they are not universally enabled by default.

C. Real World Cases of Public Wi-Fi Attacks

The threats described above are not purely theoretical. There are several documented incidents that demonstrate the real-world impacts of public Wi-Fi vulnerabilities.

In 2010, a Firefox extension called Firesheep was publicly released as a demonstration tool to show users a lack of communication encryption on websites. This tool allowed anyone on the same network to capture session cookies from sites such as Facebook and Twitter on open Wi-Fi networks without technical expertise [15]. This sparked widespread concern by showing how accessible session hijacking had become, and how the burden of protection fell on users who had no way to detect the attack. After this incident, many websites shifted to HTTPS, supporting full encryption, and Firesheep's initial attack surface disappeared.

In 2017, the discovery of the KRACK (Key Reinstallation Attack) vulnerability revealed a fundamental flaw in the WPA2 protocol. KRACK allowed attackers in a close proximity of a victim to exploit the WPA2 handshake process. The attackers replay handshake messages, which forces victims to reinstall session keys, resetting nonces and breaking the WPA's encryption [16]. Eventually, after it became known, patches were released. This was an example demonstrating that the Wi-Fi protocol didn't guarantee safety and confidentiality, and that user-side awareness remains critical.

These cases demonstrate that public Wi-Fi threats are neither theoretical nor rare. From the passive interception enabled by Firesheep to the protocol-level vulnerability exposed by KRACK, adversaries have consistently shown that public networks can be exploited with modest resources. The common theme amongst these cases is that victims had little to no awareness of the fact that an attack was taking place. Public Wi-Fi is dangerous not only because of the sophistication of attacks available to adversaries, but because the attacks themselves are often invisible to users. Without a clear understanding of what threats exist and how they operate, users remain vulnerable

regardless of their intuition about risk. This gap between the existence of threats and user awareness forms the central motivation for our study.

IV. THREAT MODEL

The threats described in Section II are not only well documented, they are achievable to carry out by adversaries with widely available technical resources against typical end users. This section provides a per-attack proof of concept, to discuss for each of the five attacks tested in our study the tools that make the attack possible, how realistically feasible it is to execute, and the reasons why a typical public Wi-Fi user is unlikely to detect or resist the attack.

We consider an opportunistic and non-targeted adversary operating within the wireless range of a public Wi-Fi network, for example, a laptop user working in a coffee shop or parked outside a hotel. The attacker's capabilities include a commodity laptop that runs penetration-testing Linux distribution such as Kali, a wireless card that supports monitor mode and packet injection, and freely available open-source tools. We assume they are of moderate but not expert skill. The adversary delivers the attack at link and network layers, either by running rogue access points within range of the real network or by associating to the real network to attack other clients. The impacts caused by the adversary are credential theft, session hijacking, redirection to malicious content and unauthorized access to services that victims use during their session. The attacks that motivate this study are those that any determined individual can execute against the general public at a commercial hotspot.

Evil Twin

Evil twin attacks are among the most accessible public Wi-Fi attacks to execute. By using open-source tools such as Wifiphisher and airbase-ng, it allows an attacker to configure rogue access points with a chosen SSID [6, 17]. These tools are preinstalled on penetration-testing distributions such as Kali Linux, with tutorials for use openly published. This makes users particularly vulnerable to ET attacks since most consumer devices automatically connect to the strongest network carrying a familiar SSID, with no built-in mechanism for users to verify that the access point is operated by a party they trust. Therefore a user walking into a coffee shop and wanting to connect to a network named "Coffee_Shop_Guest" has no reliable way to distinguish a real hotspot from a rogue one setup nearby.

Packet Sniffing

Carrying out packet sniffing attacks require no active interaction with the network and can be accomplished by adversaries using widely available free tools such as Wireshark and tcpdump. The release of Firesheep in 2010 demonstrated this threat at the lowest possible skill level because it allowed anyone on an open Wi-Fi network to capture session cookies without any technical expertise [15]. Although widespread adoption of HTTPS reduced the surface area for this attack, packet sniffing remains effective against any traffic that escapes encryption, which includes DNS queries and content delivered over any remaining HTTP endpoints [18].

Man-In-The-Middle Attacks (MITM)

MITM attacks on public Wi-Fi can be carried out using tools such as Ettercap, which automate the underlying mechanism ARP spoofing. In ARP spoofing, the attacker sends forged ARP replies to poison the victim's and gateway's ARP caches, causing both to route traffic through the attacker's machine. Then the attacker forwards packets transparently while recording them [11, 19]. Like ET attacks, such tools come preinstalled on common penetration testing distributions and are widely documented. From the user's point of view, websites continue to load and connections appear to be normal, the attack provides no visible indicator. While HTTPS protects against passive observation, MITM can be combined with other attack techniques to capture sensitive data even from sites that use HTTPS.

SSL Stripping

SSL stripping is a variation of the MITM attack first demonstrated in 2009, using the publicly released sslstrip tool which transparently downgrades HTTPS requests to HTTP without resulting in any browser warnings for the user [9]. This tool remains functional against websites that do not enforce HSTS (HTTP Strict Transport Security) modern browsers with the HSTS preload list [10]. Modern equivalents such as bettercap and SSLsplit, continue to make this attack practical [20]. Everyday users are particularly vulnerable to SSL Stripping because most do not actively check whether the browser address bar contains "https://" versus "http://" and many do not understand the difference [3, 20]. As long as the website appears to load normally, users feel no reason to suspect that anything is wrong and that their data is being transmitted in plaintext to the attacker. On public Wi-Fi, where the attacker is already well-positioned to intercept traffic, SSL stripping converts an encrypted session into a fully observable one without any action from the victim.

DNS Spoofing

DNS Spoofing on public Wi-Fi can be executed using tools such as dnsspoof and dnscraf as they intercept DNS queries and return IP address that the attacker can control [12]. This technique is especially effective combined with ET attacks. A rogue access point is by design the DHCP and DNS server for any client that associates to it, therefore it can answer DNS queries with whichever addresses it chooses, thus no active interception is required. Even on real networks, an attacker positioned between the user and the real DNS resolver can inject forged responses. The user is redirected to the malicious destination where it closely mimics the appearance of the intended site to capture credentials or deliver malware [21]. Everyday users have no practical way to detect DNS spoofing in the moment since the URL may appear correct and only with careful inspection of certificate details or network behaviour would reveal that the user had been redirected to another site. Users are not trained to perform such inspections during routine online browsing.

These five attacks share three properties that frame our study. Firstly, each attack can be achieved with freely available tools by attackers of modest skill which makes them realistic threats instead of theoretical concerns. Secondly, each attack is largely or completely invisible to the victim during execution, which means that even users who are aware of public Wi-Fi

attacks may have limited natural defenses. Thirdly, each attack succeeds by exploiting reasonable user expectations about how browsers and networks behave. Altogether, these properties motivate the three research questions introduced in Section I and shape the scenario-based methodology described in Section V.

V. METHODOLOGY

A. Study Design Overview

Our study design uses a quantitative survey-based approach aimed at measuring public Wi-Fi awareness among users. This research is structured around a two-part survey deployed through SurveyMonkey. Part I of the survey builds a demographic and behavioural baseline for each participant. Part II of the survey assess their practical awareness of public Wi-Fi threats through scenario-based questions. Together, Part I and Part II of the survey allows for both descriptive analysis of awareness levels and correlation analysis between participant characteristics and threat recognition. Our study measures whether participants can recognize and respond to attacks in practice, not determine whether they have heard of them or not.

B. Participant Recruitment and Ethics

Participants were recruited through convenience sampling, where our research team reached out to individuals within our personal and academic networks. The sample is not intended to be fully representative of the general population since recruitment relied on personal networks. However, efforts were made so that the sample included participants from a range of technical backgrounds which includes technology, business, healthcare, education, arts and trades as well as varied age groups and levels of public Wi-Fi usage. There were no incentives offered for participation. Participants consented to the study by reading the consent form and explicitly answering the first question to agree to consent prior to beginning Part I of the survey. A total of 39 participants completed both Part I and Part II of our survey.

Upon completing Part II of the survey, the participants can voluntarily provide an email address to receive a follow-up report containing their individual overall awareness score, awareness tier classification and educational material explaining the public Wi-Fi threats tested in each scenario. The educational material contains an explanation of how the attacks work, warning signs to pay attention to, and what protective measures are effective. We included this debrief to ensure that participants in the study can receive a direct educational benefit which contributes to the goal of improving public Wi-Fi awareness.

C. Survey Structure

The first survey collected demographic and behavioural baseline information across six different sections. The first section gathered demographic information including age range, field of study or work and level of technology involvement in their professional or academic life. The second section assessed self-reported technical knowledge and Wi-Fi familiarity using five-point Likert scales. The third section examined public Wi-Fi usage habits, including frequency of use, typical locations, devices used and activities done on public networks. The fourth

section explored participants' perceptions and emotional attitudes toward public Wi-Fi security, including a question on concern level (five-point scale) and general Wi-Fi security attitude, both of which were designed to enable optimism bias analysis. The fifth section assessed baseline security practices such as VPN awareness and usage, attention to HTTP versus HTTPS and whether participants read terms of use or captive portal agreements. The final section directly asked participants whether they were aware of any public Wi-Fi security threats, both through open-ended response and by selecting specific attack types they had heard of by name (MITM, evil twin, packet sniffing, DNS spoofing, and SSL stripping).

For Part II of the survey, we employed a scenario-based approach to measure a user's practical awareness of public Wi-Fi threats. Each scenario described a realistic situation without naming the exact attack type. This way the survey prevents participants from being influenced to select answer options containing technical terms that could overstate their actual level of understanding.

Each scenario is set in a distinct real-world location: a coffee shop (evil twin), an airport (passive sniffing), a library (MITM), a hotel (SSL stripping) and a café (DNS spoofing). Within each scenario, four questions are presented to assess the participants' behavioural response, ability to identify a threat, understanding of how the current attack tested works and their perception of risk. Scenarios 1, 2, 3, and 5 describe situations with open networks. The fourth scenario (SSL stripping) was deliberately set on a password-protected network. We use this design choice to test whether participants show a false sense of security from using a password protected Wi-Fi network. Additionally, survey questions were presented one at a time to prevent participants from adjusting previous answers based on later questions.

D. Scoring and Classification

The responses from Part II of the survey were scored using a 0-2 point rubric which is applied to each of the four questions within each scenario. A score of 0 indicates no awareness of the threat or a false sense of security. A score of 1 indicates partial awareness, where the participant has some intuition that there is something wrong but demonstrates either an incomplete understanding or inconsistent protective behaviour. Achieving the maximum score of 2 indicates full awareness, where the participant correctly identifies the threat or would take an appropriate protective action.

Most of the Part II survey questions are single-select, such that the participant chooses one response and the score is determined directly in the rubric. There are two questions (Scenario 2 Question B and Scenario 3 Question C) that use a multi-select format with modified scoring rules. For Scenario 2 Question B, correct selections each contribute to a maximum of two points; if a participant selects correct items alongside incorrect responses, the score is capped at 1 regardless of how many correct responses were selected. In Scenario 3 Question C, the same additive scoring applies, except incorrect selections each subtract one point from the total with the minimum floored at 0. These two scoring rules are intentionally different since the first tests recognition of what is visible on a network, where

selecting an incorrect overestimation alongside correct items suggest incomplete understanding. The second tests knowledge of protective measures, where selecting ineffective protections along with effective ones suggests confusion about which measure or tools help.

There are four questions per scenario and five scenarios in total, thus the maximum possible score is 40 points. Based in their total score, participants are classified into four overall awareness tiers. Scores between 0 to 10 indicate an “Unaware” participant who can’t recognize threats and takes no precautions. Scores between 11 to 20 indicate a “Partially Aware” participant with some intuition but significant gaps in understanding or behaviour. Scores between 21 to 30 indicates a “Mostly Aware” participant who can recognize threats and take some precautions, though the participant may not fully understand the underlying mechanisms. Scores between 31 to 40 indicate a “Fully Aware” participant who identifies the threats, understands why they are dangerous and responds appropriately. Additionally, these tier thresholds are applied at the individual scenario level on a 0 to 8 scale; Unaware: 0 to 2, Partially Aware: 3 to 4, Mostly Aware: 5 to 6, Fully Aware: 7 to 8. So for each scenario, a participant obtains a score out of 8 which allows for comparisons of awareness across attack types.

E. Optimism Bias Measurement

We use two questions from Part I of the survey to assess optimism bias. We compared the participants’ self-reported security concern and Wi-Fi security attitude. And so, a participant is classified as exhibiting optimism bias if they meet all three points of the following criteria:

1. A security concern rating of 1 to 2 on the five-point scale indicating low concern.
2. A Wi-Fi security attitude of “Don’t think about it” or “Assume it’s mostly safe”.
3. Scoring in the Unaware or Partially Aware tier.

We considered a more general definition of optimism bias where a participant’s response could either have the first or second option paired with scoring 20 points or less. However, we deemed the first definition to be more methodologically sound for this study since it requires two self-reported measures that are independent of each other to occur before classifying a participant as showing optimism bias.

F. Statistical Analysis

To evaluate our participant’s results, all statistical analyses were conducted using Python with the NumPy, pandas, SciPy, and matplotlib in Google Colab. Additionally, we selected non-parametric tests because of the ordinal nature of the self-report scales and the small sample size ($n=39$), which does not satisfy the normality assumptions which is required by parametric alternatives.

We used Spearman rank correlations to assess relationships between ordinal self-report variables such as technical knowledge and security concern as well as awareness scores. Mann-Whitney U tests were used to analyze comparisons between two independent groups such as participants who work in technology versus those who do not. Kruskal-Wallis tests were used for comparisons across three or more groups such as

self-reported threat awareness (Yes/Somewhat/No). The Friedman test was used to compare awareness scores across the five attack scenarios, since they represent repeated measures from the same participants. For cases when the Friedman test results were significant, we conducted post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction to identify which specific scenario pairs differed significantly.

VI. RESULTS

This paper’s section presents the results of our survey study which are organized by our three research questions. Our study included 39 participants who completed both parts of the survey. Table I summarizes participant demographics, Table II presents the per-scenario awareness scores, and Table III reports the results of the statistical tests done on the survey results.

Table 1: Participant Demographics

Category	Group	N	%
Age	18-24	13	33.3
	25-34	17	43.6
	35-44	3	7.7
	45-54	2	5.1
	55+	4	10.3
Field of study/work	Tech/IT/SW/Eng	12	30.8
	Business/Finance	7	17.9
	Trades/Labour	5	12.8
	Arts/Media/Design	5	12.8
	Education	3	7.7
	Healthcare	2	5.1
	Other	4	10.3
Tech Experience	Direct Tech	12	30.8
	Regular User	21	53.8
	Occasional	4	10.3
	Minimal	1	2.6
Tech Knowledge (1-5)	1 (Very Low)	1	2.6
	2 (Low)	5	12.8
	3 (Moderate)	16	41.0
	4 (High)	14	35.9
	5 (Very High)	3	35.9
Wi-Fi Usage Frequency	Never	2	7.7
	Rarely	9	5.1
	Sometimes	14	23.1
	Frequently	14	35.9
Security Concern (1-5)	1 (Not at all)	5	12.8
	2 (Not so concerned)	12	30.8
	3 (Somewhat)	17	43.6
	4 (Very)	3	7.7
	5 (Extremely)	2	5.1

RQ1: Overall Awareness and Attack Specific Blind Spots

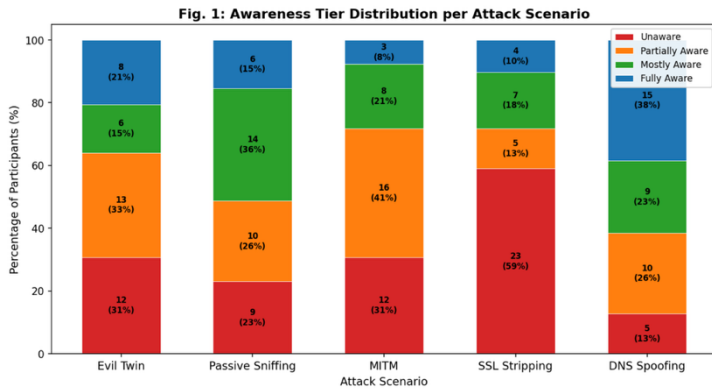
The average overall awareness tier across all participants is Partially Aware (mean score=19.9/40, median=18, standard deviation=8.54). Most participants fell into the Partially Aware tier, where nearly half ($n=19$, 49%) of the participants are classified as Partially Aware, 28% ($n=11$) are classified as Mostly Aware, and only 13% ($n=5$) fell into the Fully Aware tier. Four participants (10%) scored in the Unaware tier. These results suggest that most public Wi-Fi users in our sample recognize that there are some risks that exist but lack the

specific knowledge needed to identify or respond accordingly to attacks.

Table 2: Per-Scenario Awareness Scores and Tier Distributions

Scenario	Mean (out of 8)	SD	# Unaware Participants	# Fully Aware Participants
Evil Twin	4.05	2.38	12	8
Passive Sniffing	4.21	2.24	9	6
MITM	3.67	1.91	12	3
SSL Stripping	2.72	2.49	23	4
DNS Spoofing	5.31	2.31	5	15
Overall Score (out of 40)	19.95	8.54	4	5

Awareness of a Wi-Fi threat was not consistent amongst attack types. The Friedman test confirmed that the awareness scores differed significantly across the five scenarios (Chi-squared=40.60 with $p<0.001$), which indicates that participant's ability to recognize threats varies depending on the type of attack (Fig. 1). For identifying which specific scenarios differed, we conducted a post-hoc pairwise signed-rank tests with Bonferroni correction with an adjusted significance threshold=0.005.



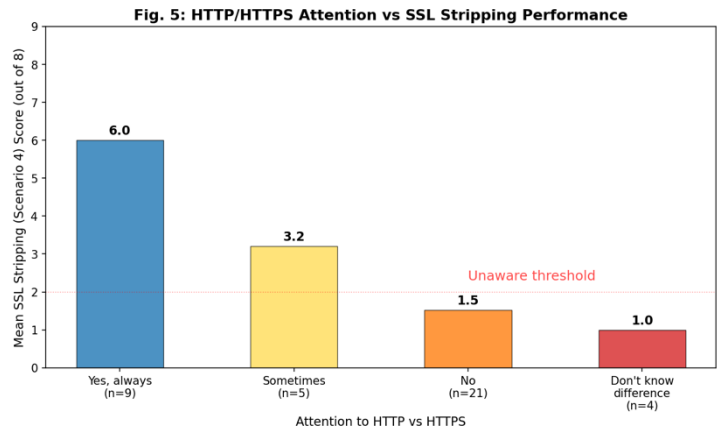
Out of the five attack scenarios, participants were most aware of DNS spoofing, with the highest mean score (5.31/8), where 38% (n=15) of the participants were classified as Fully Aware. On the contrary, SSL Stripping had the lowest mean score of 2.7/8, where 59% (n=23) of participants were classified as Unaware. The 2.6 points on our 8-point scale gap in awareness between these two scenarios shows the difference between a participant who can identify and respond to a threat and one who does not recognize that the threat exists. Five pairwise comparisons that survived the Bonferroni correction are:

- SSL Stripping vs DNS Spoofing ($p<0.001$)
- MITM vs. DNS Spoofing ($p<0.001$)
- Passive Sniffing vs. SSL Stripping ($p<0.001$)
- Evil Twin vs. SSL Stripping ($p=0.001$)

- Evil Twin vs. DNS Spoofing ($p=0.003$)

The Evil Twin, Passive Sniffing, and MITM scenarios did not significantly differ from each other, with means between 3.67 and 4.21 out of 8, all within the Partially Aware tier.

From our sample, 23% (n=9) of participants reported that they always check for HTTPS, and their mean SSL Stripping score was 6.0/8, placing them in the Mostly Aware tier for this scenario. The 54% (n=21) who reported never paying attention to HTTPS scored a mean of 1.5/8 on SSL Stripping and the 10% (n=4) of participants who did not know the difference between HTTPS and HTTP scored 1.0/8, both groups fell within the Unaware tier for SSL Stripping (Fig. 5).



To determine if Wi-Fi threat awareness is consistent across scenarios for each participant, we calculated the average cross-scenario standard deviation which is 1.82 points, thus indicating a moderate variability of scenario awareness scores within each individual participant.

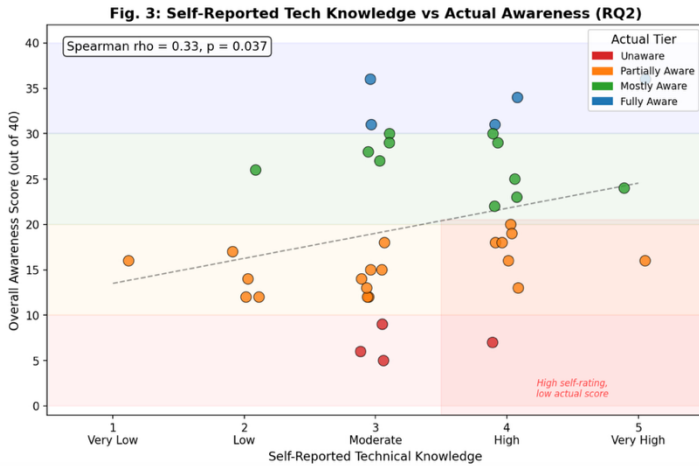
In terms of their self-reported familiarity with attack terminology from Part 1 of the survey, 62% (n=24) participants had not heard of any of the five attack types by name. MITM and DNS Spoofing were tied for the highest recognition at 31% each, whereas Evil Twin and SSL Stripping were recognized by only 10% of participants. Additionally, participants who have heard of a given attack consistently scored higher on its corresponding scenario. For instance, those who had heard of MITM scored a mean score of 5.1/8, falling under Mostly Aware in comparison to those who have not at 3.0/8, falling under the Partially Aware tier. Overall, the number of attack types a participant had heard of was the strongest predictor of overall awareness in this study (Spearman rho=+0.697 with $p<0.001$), which is stronger than if the participant had self-reported that they had technical knowledge, Wi-Fi familiarity or security concern level (Table III, Fig. 2).

RQ2: Self-reported technical knowledge vs. actual awareness

Self-reported technical knowledge was positively correlated with actual Wi-Fi threat awareness (Spearman rho=+0.334, $p=0.037$) which confirms that participants in our sample who consider themselves more technically knowledgeable do tend

to score higher. However, the practical significance of this relationship is limited. The participants that rated themselves as having High or Very High ($n=17$) technical knowledge on average were classified as Mostly Aware (mean score=22.4/40), while those who rated themselves lower ($n=22$) averaged as Partially Aware (mean score=18.0/40). This 4.4-point gap did not reach statistical significance ($U=254.5$, $p=0.057$), though the overall correlation between technology knowledge and awareness remains significant.

The more revealing result from our study is the proportion of self-described tech-savvy participants who scored poorly. Of the 17 participants who rated their technical knowledge as High or Very High, 47% ($n=8$) scored at or below the Partially Aware tier. Thus, showing that although self-reported technical knowledge is positively correlated with awareness, it is not guaranteed (Fig. 3).



SSL Stripping proved to be the most challenging for tech-savvy participants, who averaged 3.2/8, placing them in the Partially Aware tier. Even the participants who are comfortable with technology may not understand the specific mechanisms of connection security protocols.

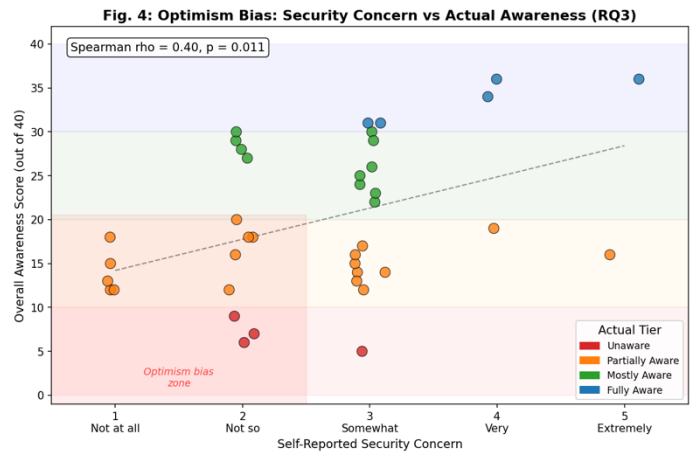
Participants who work or study directly in technology ($n=12$) scored significantly higher overall (mean=27.5/40) than those who do not ($n=27$, mean=16.6/40, $U=283.5$, $p<0.001$). This 10.9-point gap spans approximately two tiers on our classification scale. Within the technology group, 50% ($n=6$) scored in the Mostly Aware tier and 25% ($n=3$) reached Fully Aware, with no technology workers falling into the Unaware tier. However, 25% ($n=3$) still fell into the Partially Aware tier. Amongst this group SSL Stripping is the weakest scenario (mean=4.5/8). In comparison, participants in Business and Finance ($n=7$, mean=14.6/40) and Trades and Labour ($n=5$, mean=10.6/40) scored considerably lower, where Trades and Labour participants averaging just above the Unaware threshold [22].

While field of study and direct technology experience indicate technical background, we also examined whether participant's own self-assessment of their threat awareness was a better predictor of actual performance. Self-reported threat awareness aligned with actual performance more closely than self-reported technical knowledge did. Participants claiming to

be aware of public Wi-Fi threats ($n=14$) scored a mean of 26.1/40, compared to 18.6/40 for those selecting "Somewhat" aware ($n=14$), and 13.8/40 for those who answered "No" ($n=11$). This decline in score is statistically significant (Kruskal-Wallis $H=14.32$, $p=0.001$). That said, awareness was not universal even among those who claimed it: four of fourteen participants who answered 'Yes' still scored in the Partially Aware tier, suggesting that users can overestimate their understanding of specific threats even when they have a general sense that risks exist.

RQ3: Optimism Bias

The participant's security concern level showed a significant positive correlation with actual awareness (Spearman $\rho=+0.405$, $p=0.011$), which means that those who expressed greater concern about public Wi-Fi security tended to perform better (Fig. 4). However, a substantial proportion exhibited what we characterize as optimism bias: low concern about the security of public Wi-Fi despite demonstrating the limited ability to identify the threats it poses.



Among the ten participants who exhibit optimism bias, they averaged in the Partially Aware tier (12.3/40). Three of these participants scored in the Unaware tier. These participants face the greatest risk as they are unable to recognize common threats and see no reason to take precautions.

Participants with low security concern on average classified as Partially Aware (mean score=17.1/40) in comparison to Mostly Aware (mean score=28.2/40) for those with a high level of concern. This 11.1-point gap is statistically significant ($U=13.5$, $p=0.025$). Although the high concern group represented 13% of the sample ($n=5$), which limits the reliability of this comparison. The Wi-Fi Security attitude reinforced this pattern. Participants who reported "Don't think about it" ($n=4$, mean=15.2/40) and those who "Assume it's mostly safe" ($n=13$, mean=17.5/40), both fell into the Partially Aware tier.

SSL Stripping provided a focused representation of how false confidence can manifest in practice, as it was deliberately set in a hotel with password protected Wi-Fi. In the final question of this scenario, the participants were told that they were about to enter their payment information on a checkout

page whose URL had switched from HTTPS to HTTP while connected to the hotel's password-protected Wi-Fi. When asked what they would do, 10% ($n=4$) chose to proceed with the transaction specifically because they believed the hotel's password made their connection secure. These participants scored a mean of 0.2/8 in this scenario in comparison to 3.0/8 for all other participants. Although this subsample is too small for statistical testing, this shows the mechanism by which password protected Wi-Fi can create a false sense of security, where participants who trusted the password saw no reason to verify the integrity of their connection [23].

Additional Observations:

We observed several additional findings that will provide context for the discussion that follows. Firstly, VPN knowledge is significantly associated with threat awareness ($U=220.0$, $p=0.005$). Those who knew what a VPN is ($n=30$) on average classified as Mostly Aware (mean score=22.1/40) in comparison to those who did not ($n=9$, average classification=Partially Aware). However, 40% ($n=12$) of participants who knew what a VPN is reported never using one on public Wi-Fi, placing them in the Partially Aware tier (mean score=18.6/40).

Having heard or experienced a negative experience related to public Wi-Fi did not significantly predict awareness scores ($U=208.0$, $p=0.427$). There were 38% ($n=15$) who had experienced it (mean score=21.2/40) and 62% ($n=24$) who did not (mean score=19.2/40). Similarly, public Wi-Fi usage frequency was not significantly associated with awareness (Kruskal-Wallis $H=3.25$, $p=0.355$).

Lastly, 23% ($n=9$) of participants were identified as high risk, meaning they reported using public Wi-Fi sometimes or frequently, performed activities such as email or banking and scored in the Unaware or Partially Aware tiers.

The correlation heatmap (Fig. 6) provides a summary of the relationships amongst all the self-reported measures and actual scenario scores. The strongest correlation of overall awareness was the number of attack types a participant had heard of ($p=0.69$), followed by security concern ($p=0.40$), Wi-Fi familiarity ($p=0.44$), and self-reported technical knowledge ($p=0.33$).

VII. DISCUSSION

This section reflects on the results of our study, discusses their implications for public Wi-Fi security, acknowledges the limitations of our approach, and offers recommendations for addressing the awareness gaps we identified.

A. Interpretation of Findings

Our results display a consistent pattern: the majority of public Wi-Fi users in our sample have a general sense that public networks carry risk. However, they lack the specific knowledge needed to identify or defend against particular attacks. The overall mean score of 19.9/40 places the average participant at the boundary between Partially Aware and Mostly Aware, which reflects a state of vague intuition rather than actionable understanding. This distinction is important because

having vague awareness alone does not change behaviour; a user who "feels like public Wi-Fi might be risky" but cannot distinguish between a legitimate and a spoofed network is functionally as vulnerable as one who has never considered the risk at all.

This disparity across the five attack types revealed by our RQ1 carries important implications. SSL Stripping emerged as a critical blind spot, where 59% of the participants classified as Unaware, and its severity is compounded by two factors. Firstly, SSL stripping targets HTTPS protocol, which is the mechanism that users rely on most for connection security [24]. Secondly, this was the only scenario set on a password-protected network, and the participants who trusted the password saw no reason to verify the integrity of their connection. These factors reinforce each other rather than being independent gaps. A user who believes a password-protected network is inherently safe has no incentive to check whether their connection has been downgraded from HTTPS to HTTP. The four participants who explicitly chose to complete their transaction because they trusted the hotel's password averaged 0.2/8 on this scenario, compared to 3.0/8 for all other participants, which illustrates the specific mechanism by which a false sense of security translates into vulnerable behaviour [25].

The SSL stripping scenario showed the greatest performance gap within our subsamples. The range from an average score of 6.0/8 ($n=9$, participants who always check for HTTPS) down to 1.0/8 ($n=4$, participants who do not know the difference between HTTPS and HTTP) suggests that awareness of the HTTPS indicator is a strong practical indicator of whether a user can identify a connection downgrade attack. Participants who reported always checking for HTTPS were the only group to reach the Mostly Aware tier in this scenario, while those who did not know the difference between HTTPS and HTTP scored at the bottom of the unaware tier. This gradient reinforces the argument that SSL stripping awareness is not an abstract technical skill but instead a concrete behavioural habit that can be taught.

The strong performance on the DNS Spoofing scenario warrants careful interpretation. We believe this result reflects the scenario design rather than a genuine technical understanding of DNS resolution. Since the scenario described visible cues such as buttons in unfamiliar positions and a website layout that felt "slightly off", this allowed participants to potentially rely on general intuition regarding suspicious websites. This interpretation is supported by the fact that DNS Spoofing was recognized by name by only 31% of participants, whilst the scenario performance was the highest of all five attacks. The implication is that visual-cue-based threats may be easier for users to detect intuitively, whereas protocol-level threats like SSL stripping remain invisible without specific technical education [26].

The finding that awareness is patchy within individuals, with an average cross-scenario standard deviation of 1.82, has practical implications for security education. It was common for participants to score Mostly Aware on one scenario while scoring Unaware on another, indicating that awareness is

attack-specific instead of a general trait. Providing broad advice such as “be careful on public Wi-Fi” is therefore unlikely to be effective. Users need targeted education that addresses each threat individually.

For RQ2, the moderate yet statistically significant correlation between self-reported technical knowledge and actual awareness confirms that a technical background provides some advantage, but far less than might be expected. The finding that the mean difference between self-rated tech-savvy and non-tech-savvy participants did not reach statistical significance ($p=0.057$), despite the significant overall correlation, reinforces the point that the relationship exists, but it is not strong enough to reliably distinguish groups defined by self-assessment. What is more revealing is that 47% of the self-described tech-savvy participants scored at or below the Partially Aware tier. They believe that they understand the risks, and that belief can potentially increase their vulnerability since it reduces their motivation to learn about specific threats that they have never encountered before.

Working directly in technology proved to be a stronger predictor than self-assessed knowledge, with technology workers scoring approximately two tiers higher on average than those who do not. This suggests that hands-on professional exposure to networks and security practices matters more than general confidence in one’s own technical abilities. However, even among the technology workers, SSL Stripping remained the weakest scenario (mean=4.5/8), and for self-described tech-savvy participants more broadly, the mean on this scenario was only 3.2/8, which places them in the Partially Aware tier. This indicates that neither professional tech experience nor general technical confidence automatically results in understanding of protocol-level vulnerabilities [27].

The strongest predictor we identified across all self-reported measures was the number of attack types a participant had heard of (Spearman $\rho=0.697$). This finding has a clear educational implication that simply exposing users to the names and basic mechanisms of common attacks may be one of the most accessible and effective interventions available. Participants do not need deep technical expertise to benefit from knowing that SSL stripping, evil twin networks, packet sniffing exist and how they generally work [28].

For RQ3, the presence of optimism bias in 26% of participants under our strict definition confirms that a meaningful portion of public Wi-Fi users feel secure despite being unable to recognize common threats. These individuals are in a particularly vulnerable position since their low concern reduces their likelihood of learning protective information or adopting protective behaviours. Three of the ten participants displaying optimism bias scored in the Unaware tier, which represents participants who simultaneously expressed low concern and demonstrated the lowest actual awareness in our sample.

Additionally, our VPN analysis illustrates the knowledge-to-practice gap. The 12 participants who knew what a VPN is yet reported never using one on public Wi-Fi scored a mean of 18.6/40, which is notably lower than the 22.1/40 average for VPN-knowledgeable participants overall in our sample. This

gap suggests that even when users understand that protective tools exist, they do not consistently adopt them. Awareness of a tool is not the same as understanding why that tool is necessary and without this understanding, adoption of a tool remains inconsistent.

Our non-significant results are equally informative. Encountering prior negative experience with public Wi-Fi did not significantly predict awareness ($p=0.427$), which aligns with the nature of the attacks: most public Wi-Fi attacks are invisible to the victim, so a user who has been attacked may never know it happened and therefore has nothing to learn from the event. Likewise, public Wi-Fi usage frequency was not significantly associated with awareness ($p=0.355$), which indicates that repeated exposure does not organically build an understanding of the risks. Together, these results reinforce the argument that awareness does not come organically from frequency of use or experience and that it likely requires deliberate education.

Lastly, 23% of our sample was identified as high risk, meaning that participants who use public Wi-Fi sometimes or frequently, performed sensitive activities such as logging into banking or email, and scored in the Unaware or Partially Aware tier. This composite is important since it identifies the population most directly vulnerable to the attacks outlined in this study. These participants are not people who occasionally browse on Wi-Fi, instead they are those who routinely expose themselves to the exact threats they cannot recognize.

B. Implications and recommendations

Based on our results, we have identified several practical implications for users, educators and network operators.

For users, the most actionable recommendation is to verify HTTPS connections on each website they browse where sensitive information is entered, regardless of whether or not the network is password protected. Our results show that attention to HTTPS is strongly associated with awareness of SSL stripping, the attack type that posed the greatest challenge for our sample. Public Wi-Fi users should also understand that an internet connection that is working does not indicate a safe connection, and that a password-protected network does not prevent interception of individual user traffic [29].

For educators and institutions, our findings suggest that security awareness programs should educate beyond general warnings about public Wi-Fi and instead use scenario-based education to address specific attack types individually. The patchy nature of awareness within individuals demonstrates that a user who understands one threat may be completely unaware of another. Programs for security awareness could potentially be structured around the five attack categories used in this study, utilizing practical examples of how each attack works and what protective measures are effective against each. Our finding that familiarity with attack terminology was the strongest predictor of awareness suggests that even brief lessons focused on terminology covering the names and basic mechanism of common attacks could produce meaningful improvements, particularly among users currently having no exposure to these terms.

Lastly, for network operators that provide password-protected public Wi-Fi, our results highlight a specific concern that passwords may create a false sense of security amongst users. Network operators can consider displaying clear warnings on captive portals or login screens that a network password does not protect individual connections from interception.

C. Limitations

There are several limitations that should be considered when interpreting our results.

Our sample of 39 participants was recruited through convenience sampling from personal and academic networks, therefore limiting the generalizability of our findings. Certain demographic groups are underrepresented, specifically in the 35-44, 45-54 and 55+ age ranges where the group sizes are too small for reliable comparisons. Furthermore, 31% of our sample also works directly in technology which is likely higher than the general population and may inflate the overall awareness scores relative to a more representative sample.

Even though our scoring rubric was designed to be systemic, it involves judgement in assigning point values to specific responses. Alternative rubrics could produce different score distributions, though the relative ordering of scenarios and the overall patterns we identified are likely to remain consistent.

Lastly, with 39 participants and multiple statistical comparisons conducted, we acknowledge the possibility of inflated Type I error. We have reported all results, including non-significant ones and applied Bonferroni correction to pairwise scenario comparisons to mitigate this concern. Group comparisons with small subsamples, such as the high concern group (n=5) and the participants who had heard of SSL Stripping (n=4), should be interpreted with caution.

D. Future Work

Our study identifies several directions for future research. Firstly, a larger and more demographically diverse sample would allow for more robust subgroup analyses, specifically across age groups and non-technical professional fields where our sample sizes were limited. A longitudinal design could test whether targeted security education structured around the five attack categories used in this study can produce lasting improvements in awareness. To bridge the gap between self-reported and actual behaviour, researchers can complement survey-based measurements with behavioural observation. Finally, the result that awareness is patchy across attack types suggests that adaptive assessment tools that identify a user's specific blind spots and provide targeted education may be more effective than general security training programs.

VIII. CONCLUSION

In conclusion, we investigated public Wi-Fi attack awareness through a two-part scenario-based survey of 39 participants across five attack types. We discovered that

awareness is attack-specific rather than general, and that if a user has prior familiarity with a specific attack's terminology, that is the strongest predictor of recognizing the attack, which outweighs their security concern or self-reported technical knowledge. We also identified approximately one quarter of our participants as showing optimism bias. These findings indicate that to close the public Wi-Fi awareness gap, general reminders to "be careful" are not sufficient and that meaningful impact instead comes from targeted education which names specific attacks and explains how they work.

REFERENCES

- [1] Kim, Young-Long. "Seoul's Wi-Fi Hotspots: Wi-Fi Access Points as an Indicator of Urban Vitality." *Computers, Environment and Urban Systems*, vol. 72, Nov. 2018, pp. 13–24, <https://doi.org/10.1016/j.compenvurbsys.2018.06.004>.
- [2] Mandal, Prianka, et al. "Free Wi-Fi Is Not Ultimately Free": Privacy Perceptions of Users in the US Regarding City-Wide Wi-Fi Services." *Proceedings on Privacy Enhancing Technologies*, 2025, <https://petssymposium.org/popets/2025/popets-2025-0111.php>.
- [3] Reese, Ken, et al. "A Usability Study of Five {Two-Factor} Authentication Methods." *Usenix.org*, 2019, pp. 357–370, www.usenix.org/conference/soups2019/presentation/reese?utm_source.
- [4] CERT-EU Security Advisory. KRACK -Key Reinstallation Attacks: Breaking WPA2. 17 Oct. 2017.
- [5] Sangeen, Muhammad, et al. "Blind-Trust: Raising Awareness of the Dangers of Using Unsecured Public Wi-Fi Networks." *Computer Communications*, vol. 209, no. 209, 1 Sept. 2023, pp. 359–367, [www.sciencedirect.com/science/article/abs/pii/S0140366423002396](https://doi.org/10.1016/j.comcom.2023.07.011), <https://doi.org/10.1016/j.comcom.2023.07.011>.
- [6] Yang, Chao, et al. "Active User-Side Evil Twin Access Point Detection Using Statistical Techniques." *IEEE Transactions on Information Forensics and Security*, vol. 7, no. 5, Oct. 2012, pp. 1638–1651, <https://doi.org/10.1109/tifs.2012.2207383>.
- [7] Hyo Jung Jin, et al. "Enhancing Sniffing Detection in IoT Home Wi-Fi Networks: An Ensemble Learning Approach with Network Monitoring System (NMS)." *IEEE Access*, 1 Jan. 2024, pp. 1–1, <https://doi.org/10.1109/access.2024.3416095>.
- [8] Sombatruang, Nissy, et al. "The Continued Risks of Unsecured Public Wi-Fi and Why Users Keep Using It: Evidence from Japan." *IEEE Xplore*, 1 Aug. 2018, ieeexplore.ieee.org/abstract/document/8514208/.
- [9] Zhraw, Duaa Sameer, et al. "Chronological Review of MITM Attacks: Challenges, Solutions and Recommendations." *Lecture Notes in Networks and Systems*, 2025, pp. 202–220, https://doi.org/10.1007/978-3-032-03406-9_13.
- [10] Siewert, Hendrik, et al. "On the Security of Parsing Security-Relevant HTTP Headers in Modern Browsers." *2022 IEEE Security and Privacy Workshops (SPW)*, May 2022, pp. 342–352, ieeexplore.ieee.org/abstract/document/9833880/, <https://doi.org/10.1109/spw54247.2022.9833880>.
- [11] Gangan, Subodh. A Review of Man-In-The-Middle Attacks. Apr. 2015.
- [12] Cloudflare. "What Is DNS Cache Poisoning? | DNS Spoofing | Cloudflare." *Cloudflare*, 2023, www.cloudflare.com/learning/dns/dns-cache-poisoning/.
- [13] Hu, Zi, et al. "Specification for DNS over Transport Layer Security (TLS)." *RFC*, 2016, www.semanticscholar.org/paper/Specification-for-DNS-over-Transport-Layer-Security-Hu-Zhu/44e318283837b7f8fa33d097b86ec7c0beadc821, <https://doi.org/10.17487/RFC7858>.
- [14] Hoffman, P., and P. McManus. "DNS Queries over HTTPS (DoH)." *Www.rfc-Editor.org*, 1 Oct. 2018, www.rfc-editor.org/rfc/rfc8484.html, <https://doi.org/10.17487/RFC8484>.
- [15] Schoen, Seth. "The Message of Firesheep: Baaaad Websites, Implement Sitewide HTTPS Now!" *Electronic Frontier Foundation*, 29 Oct. 2010, www.eff.org/deeplinks/2010/10/message-firesheep-baaaad-websites-implement.

- [16] Alhamry, Mohamed, and Wael Elmedany. "Exploring Wi-Fi WPA2 KRACK Vulnerability: A Review Paper." *IEEE Xplore*, 1 Oct. 2022, ieeexplore.ieee.org/document/10041548/.
- [17] Chatzisofofroniou, George, and Panayiotis Kotzanikolaou. "Exploiting Wi-Fi Usability Features for Association Attacks in IEEE 802.11: Attack Analysis and Mitigation Controls1." *Journal of Computer Security*, 11 Apr. 2022, pp. 1–24, <https://doi.org/10.3233/jcs-210036>.
- [18] Hoang, Nguyen Phong, et al. "Assessing the Privacy Benefits of Domain Name Encryption." *Proceedings of the 15th ACM Asia Conference on Computer and Communications Security*, 5 Oct. 2020, <https://doi.org/10.1145/3320269.3384728>.
- [19] Ramachandran, Vivek, and Sukumar Nandi. "Detecting ARP Spoofing: An Active Technique." *Information Systems Security*, 2005, pp. 239–250, https://doi.org/10.1007/11593980_18.
- [20] Kim, Minju, et al. "MitM Tool Analysis for TLS Forensics." *IEEE Xplore*, 1 Aug. 2021, ieeexplore.ieee.org/abstract/document/9680752. Accessed 6 Sept. 2023.
- [21] Son, Sooel, and Vitaly Shmatikov. "The Hitchhiker's Guide to DNS Cache Poisoning." *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, 2010, pp. 466–483, https://doi.org/10.1007/978-3-642-16161-2_27.
- [22] Keshvadi, Sina. Enhancing Western Organizational Cybersecurity Resilience through Tailored Education for Non-Technical Employees. 1 Nov. 2023, pp. 1–6, <https://doi.org/10.1109/ihtc58960.2023.10508824>.
- [23] Mayer, Peter, and Melanie Volkamer. "Addressing Misconceptions about Password Security Effectively." *Proceedings of the 7th Workshop on Socio-Technical Aspects in Security and Trust - STAST '17*, 2018, <https://doi.org/10.1145/3167996.3167998>.
- [24] Hossain, Md. Shohrab, et al. "Survey of the Protection Mechanisms to the SSL-Based Session Hijacking Attacks." *Network Protocols and Algorithms*, vol. 10, no. 1, 1 Apr. 2018, p. 83, <https://doi.org/10.5296/npa.v10i1.12478>. Accessed 10 Sept. 2019.
- [25] P. Mayer and M. Volkamer, "Addressing misconceptions about password security effectively," *Proceedings of the 7th Workshop on Socio-Technical Aspects in Security and Trust*, pp. 16–27, Dec. 2018. doi:10.1145/3167996.3167998
- [26] Moreno-Fernández, María M., et al. "Fishing for Phishers. Improving Internet Users' Sensitivity to Visual Deception Cues to Prevent Electronic Fraud." *Computers in Human Behavior*, vol. 69, Apr. 2017, pp. 421–436, [dl.acm.org/citation.cfm?id=3076524](https://doi.org/10.1016/j.chb.2016.12.044), <https://doi.org/10.1016/j.chb.2016.12.044>.
- [27] Schneider, Robert. *Enhancing Information Security Literacy: The Impact of Framing across Non-Technical Trainees*. 7 Mar. 2024, www.researchgate.net/publication/378775765_Enhancing_Information_Security_Literacy_The_Impact_of_Framing_Across_Non-Technical_Trainees.
- [28] Rantos, Konstantinos, et al. "How Effective Is Your Security Awareness Program? An Evaluation Methodology." *Information Security Journal: A Global Perspective*, vol. 21, no. 6, Jan. 2012, pp. 328–345, <https://doi.org/10.1080/19393555.2012.747234>.
- [29] Maimon, David, et al. "Self-Protective Behaviors over Public Wi-Fi Networks." *Www.usenix.org*, 2017, www.usenix.org/conference/laser2017/presentation/maimon.

APPENDIX

Table 2 : All Four Tiers

Per-Scenario Awareness Scores and Tier Distributions

Scenario	Mean (out of 8)	SD	# Unaware Participants	# Partially Aware Participants	# Mostly Aware Participants	# Fully Aware Participants
Evil Twin	4.05	2.38	12	13	6	8
Passive Sniffing	4.21	2.24	9	10	14	6
MITM	3.67	1.91	12	16	8	3
SSL Stripping	2.72	2.49	23	5	7	4
DNS Spoofing	5.31	2.31	5	10	9	15
Overall Score (out of 40)	19.95	8.54	4	19	11	5

Table 3.1: Spearman Rank Correlations with Overall Score

Variable	rho	p	Sig
Self-reported tech knowledge	0.334	0.0374	*
Wi-Fi familiarity	0.443	0.0047	**
Security concern level	0.405	0.0106	*
Number of attacks heard of	0.697	0.0000	***

Table 3.2: Mann-Whitney U Tests (with rank-biserial effect size)

Comparison	n1	n2	U	p	Sig	r	Effect
Heard Evil Twin vs not	4	35	106.0	0.0965	n.s.	-0.514	Large
Heard Passive Sniffing vs not	7	32	167.5	0.0422	*	-0.496	Medium
Heard MITM vs not	12	27	257.0	0.0035	**	-0.586	Large
Heard SSL Stripping vs not	4	35	131.0	0.0045	**	-0.871	Large
Heard DNS Spoofing vs not	12	27	216.5	0.0950	n.s.	-0.336	Medium
Tech knowledge High (4–5) vs Low (1–3)	17	22	254.5	0.0574	n.s.	-0.361	Medium
Concern Low (1–2) vs High (4–5)	17	5	13.5	0.0251	*	+0.682	Large
Knows VPN vs does not	30	9	220.0	0.0048	**	-0.630	Large
Bad experience vs none	15	24	208.0	0.4267	n.s.	-0.156	Small

Table 4.1: Password-protected Wi-Fi false security

Metric	Value
Chose "secure through hotel password" on S4-QD	4 / 39 (10%)
Mean score (selected group)	0.2 / 8
Mean score (others)	3.0 / 8

Table 4.2: HTTP/HTTPS attention vs SSL stripping score

Response	n	S4 Mean (out of 8)	Overall Score (out of 40)
Yes	9	6.0	30.6
Sometimes	5	3.2	18.8
No	21	1.5	17.1
Don't know difference	4	1.0	12.5

Table 4.3: Online banking on public Wi-Fi

Metric	Value
Participants using banking	4 / 39 (10%)
Mean score	13.8 / 40
Low scorers (within group)	3 / 4

Table 4.4: Activities on public Wi-Fi

Activity	Count	Percentage
Browsing	32	82%
Social media	24	62%
Email	25	64%
Banking	4	10%
Streaming	10	26%
Work/School	20	51%
Shopping	6	15%

Table 4.5: VPN knowledge and usage

Group	n	Mean Score (out of 40)
Knows VPN (Yes)	30	22.1
Does not know VPN	4	13.8
Heard of VPN	5	12.2
Knows but never uses	12	18.6

Table 4.6: Bad experience vs awareness

Group	n	Mean Score (out of 40)
Yes	15	21.2
No	24	19.2

Table 4.7: High-risk participants (frequent + sensitive activity + low score)

Metric	Value
Count	9 / 39
Percentage	23%

Table 4.8: Cross-Scenario consistency

Metric	Value
Average SD	1.82
Interpretation	Awareness is patchy across attack types

Fig. 2: Attack Recognition and Its Effect on Scenario Performance

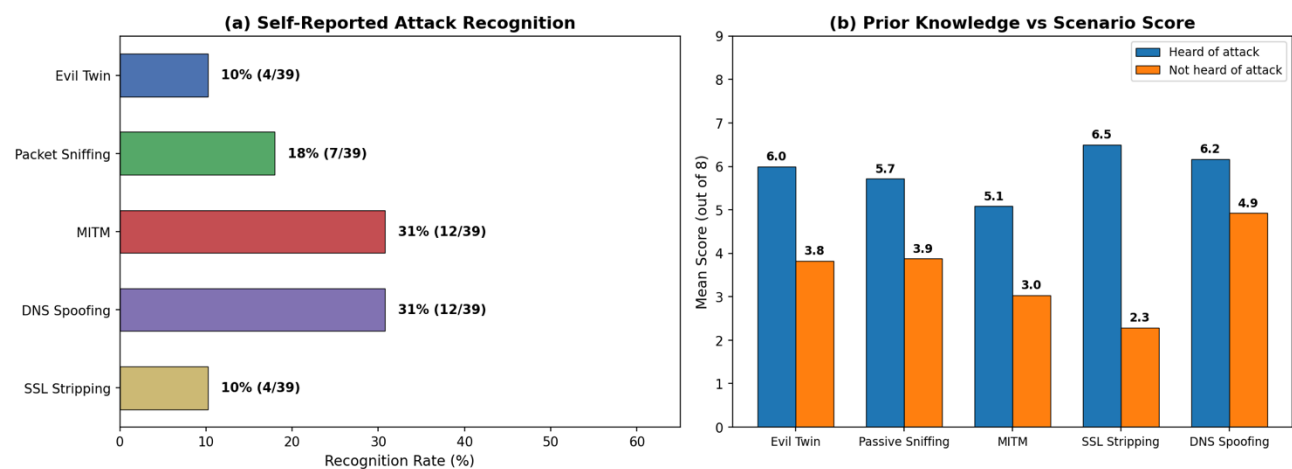
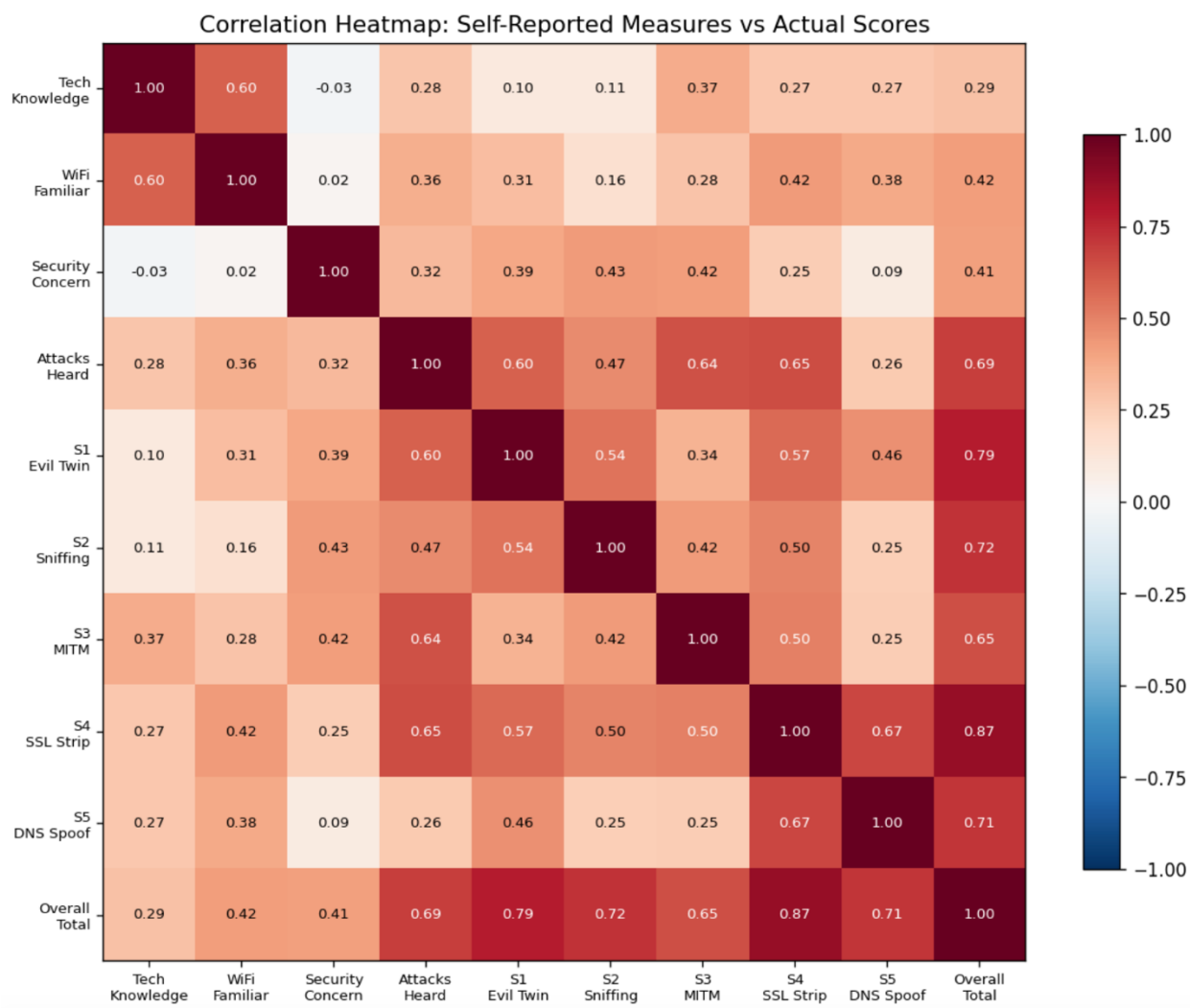


Figure 6:



Additional Survey Data Analyses:

User Awareness Per Scenario

--- Evil Twin (S1) ---

Mean: 4.05/8 | Median: 4.0

Tier distribution:

Unaware: 12 (30.8%)

Partially Aware: 13 (33.3%)

Mostly Aware: 6 (15.4%)

Fully Aware: 8 (20.5%)

--- Passive Sniffing (S2) ---

Mean: 4.21/8 | Median: 5.0

Tier distribution:

Unaware: 9 (23.1%)

Partially Aware: 10 (25.6%)

Mostly Aware: 14 (35.9%)

Fully Aware: 6 (15.4%)

--- MITM (S3) ---

Mean: 3.67/8 | Median: 3.0

Tier distribution:

Unaware: 12 (30.8%)

Partially Aware: 16 (41.0%)

Mostly Aware: 8 (20.5%)

Fully Aware: 3 (7.7%)

--- SSL Stripping (S4) ---

Mean: 2.72/8 | Median: 2.0

Tier distribution:

Unaware: 23 (59.0%)

Partially Aware: 5 (12.8%)

Mostly Aware: 7 (17.9%)

Fully Aware: 4 (10.3%)

--- DNS Spoofing (S5) ---

Mean: 5.31/8 | Median: 5.0

Tier distribution:

Unaware: 5 (12.8%)

Partially Aware: 10 (25.6%)

Mostly Aware: 9 (23.1%)

Fully Aware: 15 (38.5%)

Self-Reported Tech Knowledge vs Actual Score

Tech Knowledge = 1 (Very Low) n=1

Overall mean: 16.0/40

Tech Knowledge = 2 (Low) n=5

Overall mean: 16.2/40

Tech Knowledge = 3 (Moderate) n=16

Overall mean: 18.8/40

Tech Knowledge = 4 (High) n=14

Overall mean: 21.8/40

Tech Knowledge = 5 (Very High) n=3

Overall mean: 25.3/40

Wi-Fi Familiarity vs Actual Scores

Wi-Fi Familiarity = 1 (Not at all) : n=2

Overall mean: 14.0/40

Evil Twin: 3.5/8

Passive Sniffing: 3.5/8

MITM: 2.5/8

SSL Stripping: 1.0/8

DNS Spoofing: 3.5/8

Wi-Fi Familiarity = 2 (Not so familiar) : n=9

Overall mean: 13.4/40

Evil Twin: 2.7/8

Passive Sniffing: 3.1/8

MITM: 2.8/8

SSL Stripping: 0.7/8

DNS Spoofing: 4.2/8

Wi-Fi Familiarity = 3 (Somewhat) : n=21

Overall mean: 21.8/40

Evil Twin: 4.3/8

Passive Sniffing: 4.8/8

MITM: 4.0/8

SSL Stripping: 3.3/8

DNS Spoofing: 5.4/8

Wi-Fi Familiarity = 4 (Very familiar) : n=4

Overall mean: 24.8/40

Evil Twin: 5.0/8

Passive Sniffing: 4.0/8

MITM: 3.8/8

SSL Stripping: 4.2/8

DNS Spoofing: 7.8/8

Wi-Fi Familiarity = 5 (Extremely) : n=3

Overall mean: 24.0/40

Evil Twin: 5.3/8

Passive Sniffing: 4.3/8

MITM: 4.7/8

SSL Stripping: 3.7/8

DNS Spoofing: 6.0/8

Optimism Bias vs Actual Score

Concern = 1 (Not at all) n=5

Overall mean: 14.0/40

Concern = 2 (Not so concerned) n=12

Overall mean: 18.3/40

Concern = 3 (Somewhat) n=17

Overall mean: 20.4/40

Concern = 4 (Very) n=3

Overall mean: 29.7/40

Concern = 5 (Extremely) n=2

Overall mean: 26.0/40

Wi-Fi Security Attitude vs Actual Score

'Don't think about it' n=4
Overall mean: 15.2/40
'Assume mostly safe' n=13
Overall mean: 17.5/40
'Somewhat cautious' n=17
Overall mean: 21.2/40
'Actively take precautions' n=2
Overall mean: 32.5/40
'Avoid public Wi-Fi' n=3
Overall mean: 21.7/40

Field of Work/Study vs Score

Other n=4
Overall mean: 17.8/40
Tech/IT/SW/Eng n=12
Overall mean: 27.5/40
Trades/Labour n=5
Overall mean: 10.6/40
Healthcare n=2
Overall mean: 18.0/40
Education n=3
Overall mean: 24.0/40
Arts/Media/Design n=5
Overall mean: 19.6/40
Business/Finance n=7
Overall mean: 14.6/40

Tech usage level vs. Scores

'Direct tech' n=12
Overall mean: 27.5/40
'Regular user' n=21
Overall mean: 17.5/40
'Occasional' n=4
Overall mean: 11.8/40
'Minimal' n=1
Overall mean: 17.0/40

Self-Reported Threat Awareness vs Actual Score

Check if people claiming to know threats score higher:

'Yes' n=14
Overall mean: 26.1/40
'Somewhat' n=14
Overall mean: 18.6/40
'No' n=11
Overall mean: 13.8/40

Heard of Specific Attack vs Score on Corresponding Scenario

Evil Twin:
Heard of it (n= 4): mean 6.0/8
Not heard of it (n=35): mean 3.8/8

Passive Sniffing:
Heard of it (n= 7): mean 5.7/8
Not heard of it (n=32): mean 3.9/8

MITM:
Heard of it (n=12): mean 5.1/8
Not heard of it (n=27): mean 3.0/8

SSL Stripping:
Heard of it (n= 4): mean 6.5/8
Not heard of it (n=35): mean 2.3/8

DNS Spoofing:
Heard of it (n=12): mean 6.2/8
Not heard of it (n=27): mean 4.9/8

HTTP/HTTPS Attention vs SSL Stripping Score

HTTP/HTTPS attention = 'Yes' n=9
S4 (SSL Stripping) mean: 6.0/8
Overall mean: 30.6/40
HTTP/HTTPS attention = 'Sometimes' n=5
S4 (SSL Stripping) mean: 3.2/8
Overall mean: 18.8/40
HTTP/HTTPS attention = 'No' n=21
S4 (SSL Stripping) mean: 1.5/8
Overall mean: 17.1/40
HTTP/HTTPS attention = 'Don't know difference' : n=4
S4 (SSL Stripping) mean: 1.0/8
Overall mean: 12.5/40

VPN Knowledge & Usage vs Scores

Knows VPN = 'Yes' n=30
Overall mean: 22.1/40
Knows VPN = 'No' n=4
Overall mean: 13.8/40
Knows VPN = 'Heard of it' n=5
Overall mean: 12.2/40

Public Wi-Fi Usage Frequency vs Scores

Frequency = 'Never' n=2
Overall mean: 17.0/40
Frequency = 'Rarely' n=9
Overall mean: 18.6/40
Frequency = 'Sometimes' n=14
Overall mean: 18.1/40
Frequency = 'Frequently' n=14
Overall mean: 23.1/40

Optimism Bias

Participants who feel safe/unconcerned but scored Unaware or Partially Aware:

Participant 2: Concern=2/5, Attitude='Assume mostly safe', Score=7/40 (Unaware)
 Participant 4: Concern=1/5, Attitude='Don't think about it', Score=12/40 (Partially Aware)
 Participant 5: Concern=1/5, Attitude='Assume mostly safe', Score=12/40 (Partially Aware)
 Participant 7: Concern=2/5, Attitude='Somewhat cautious', Score=18/40 (Partially Aware)
 Participant 11: Concern=2/5, Attitude='Somewhat cautious', Score=20/40 (Partially Aware)
 Participant 12: Concern=2/5, Attitude='Assume mostly safe', Score=6/40 (Unaware)
 Participant 18: Concern=2/5, Attitude='Don't think about it', Score=18/40 (Partially Aware)
 Participant 23: Concern=3/5, Attitude='Assume mostly safe', Score=12/40 (Partially Aware)
 Participant 26: Concern=2/5, Attitude='Assume mostly safe', Score=9/40 (Unaware)
 Participant 27: Concern=1/5, Attitude='Don't think about it', Score=13/40 (Partially Aware)
 Participant 33: Concern=2/5, Attitude='Assume mostly safe', Score=16/40 (Partially Aware)
 Participant 35: Concern=1/5, Attitude='Avoid public Wi-Fi', Score=15/40 (Partially Aware)
 Participant 36: Concern=1/5, Attitude='Don't think about it', Score=18/40 (Partially Aware)
 Participant 37: Concern=2/5, Attitude='Assume mostly safe', Score=12/40 (Partially Aware)
 Participant 39: Concern=3/5, Attitude='Assume mostly safe', Score=16/40 (Partially Aware)

15/39 <= show optimism bias

Are Self-Proclaimed Tech-Savvy Participants Actually Tech Savvy:

Participants with high self-reported tech knowledge (4-5) who scored $\leq 20/40$:

Participant 2: Tech Knowledge=4/5, Score=7/40 (Unaware)
 Participant 7: Tech Knowledge=4/5, Score=18/40 (Partially Aware)
 Participant 11: Tech Knowledge=4/5, Score=20/40 (Partially Aware)
 Participant 27: Tech Knowledge=4/5, Score=13/40 (Partially Aware)
 Participant 29: Tech Knowledge=4/5, Score=19/40 (Partially Aware)
 Participant 33: Tech Knowledge=5/5, Score=16/40 (Partially Aware)
 Participant 36: Tech Knowledge=4/5, Score=18/40 (Partially Aware)
 Participant 39: Tech Knowledge=4/5, Score=16/40 (Partially Aware)

⇒ 8 of 17 tech-savvy participants (47%) scored at or below Partially Aware

Age Group vs Awareness Score

Age 18-24 n=13

Overall mean: 19.0/40

Evil Twin: 3.5/8

Passive Sniffing: 4.1/8

MITM: 3.4/8

SSL Stripping: 2.5/8

DNS Spoofing: 5.5/8

Age 25-34 n=17

Overall mean: 21.6/40

Evil Twin: 4.5/8

Passive Sniffing: 4.3/8

MITM: 4.2/8

SSL Stripping: 3.0/8

DNS Spoofing: 5.6/8

Age 35-44 n=3

Overall mean: 16.0/40

Evil Twin: 3.3/8

Passive Sniffing: 2.3/8

MITM: 3.3/8

SSL Stripping: 2.3/8

DNS Spoofing: 4.7/8

Age 45-54 n=2

Overall mean: 20.5/40

Evil Twin: 5.0/8

Passive Sniffing: 4.0/8

MITM: 2.5/8

SSL Stripping: 3.5/8

DNS Spoofing: 5.5/8

Age 55+ n=4

Overall mean: 18.5/40

Evil Twin: 4.0/8

Passive Sniffing: 5.8/8

MITM: 3.2/8

SSL Stripping: 2.0/8

DNS Spoofing: 3.5/8